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Intuitive approach to algorithmic probability

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"Information theory must precede probability theory, and not be based on it. By the very essence of this discipline, the foundations of information theory have a finite combinatorial character."

*Kolmogorov, A. N. (1983). Combinatorial foundations of information theory and the calculus of probabilities.
Russian mathematical surveys, 38 (4), 29-40.*

Problem: Probability has to do with chance. How can it be defined through algorithms?

For Algorithmic Information Theory (AIT), information depends on **complexity**.
For traditional definitions of information, information depends on **probability**.

Question: Is this collision between **complexity** and **probability** more than anecdotal?

Answer: Yes. We can take advantage of it to do nothing less than redefine probability!

The axiomatic theory of probability (also due to Andrei Kolmogorov!) puts constraints on what a probability measure can be, but does not construct it. Kolmogorov complexity can be used to define a new notion of probability, called **algorithmic probability**.

The link between complexity and probability has ancient roots. The French mathematician Émile Borel wrote in 1909:

"il est clair que les éléments pour lesquels le nombre de mots nécessaires à la définition est extrêmement grand devront être regardés comme ayant une probabilité extrêmement petite."	"it is clear that elements requiring an extremely large number of words for their definition should be considered as having an extremely low probability."
(Borel E., 1909 p. 272).	

In other words, numbers that have high program-size complexity are less probable.

To formalize this idea, Ray Solomonoff and his followers considered the probability that an object will be generated by chance using a universal Turing machine when the machine is given random programs as input.

Algorithmic probability and prefix complexity



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advantages.

For instance,

it was used in the 1970s by Leonid Levin to find a consistent definition of randomness for infinite sequences.

In conclusion,

Probability can be defined in purely algorithmic terms,

by considering only the length of programs that generate objects.

But this requires a new

► Algorithmic Probability and Prefix Complexity in a nutshell

Imagine a universal Turing machine fed with random programs (one may flip a coin to determine the successive bits of each program). Let's say that the **algorithmic probability** of s is the probability of getting s as output this way. If s can be generated by one of these random programs p of length $|p|$, then p contributes by $2^{-|p|}$ to the probability of getting s . So it's tempting to say that the algorithmic probability of s is the sum of $2^{-|p|}$ over all programs p that generate s .

This way of founding algorithmic probability is illuminating. However, as explained in the video, there is a difficulty. If s can be generated using a program p , then it can be generated by any program pq_i where q_i is a continuation of p that does anything except erasing s . If we add up the probabilities of all the pq_i , the sum will not converge.

Instead of considering all programs that generate s when computing the probability of s , why not keep the only shortest one? Since the length of the shortest program is $C(s)$, we get an alternative definition of algorithmic probability:

$$P_a(s) = 2^{-C(s)}$$

By considering only one term, one solves the problem of the diverging sum. However, we still do not have a probability, as the sum of all $P_a(s)$ for different s is infinite instead of being equal to 1.

Alternative Definition of Algorithmic Probability

1/1 point (ungraded)

Let's consider only integers.
 P_a cannot be a probability, because:

- ☒ $P_a(n)$ is close to $\frac{1}{n}$
- ☐ $P_a(n)$ is close to $\frac{1}{\log_2(n)}$
- ☐ $P_a(n)$ is close to $\log_2(n)$
- ☐ $P_a(n)$ is close to $\frac{1}{1+\log_2(n)}$

✓
Answer
Correct:
 $P_a(n)$ is close to $\frac{1}{n}$. As a consequence, the sum of the $P_a(n)$ over all integers diverges, instead of being equal to one.

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